



Non-*Saccharomyces* yeasts from fermented grains of *Baijiu* for low alcoholic beer brewing and their impacts on beer flavor

Jinjing Wang^{a,*}, Xinyue Gu^a, Glen Fox^b, Feiyun Zheng^a, Chunfeng Liu^a, Chengtuo Niu^a, Qi Li^a

^a Key Laboratory of Industrial Biotechnology, Ministry of Education, School of Biotechnology, Jiangnan University, Wuxi 214122, Jiangsu, China

^b Department of Food Science and Technology, University of California, Davis, CA 95616, United States

ARTICLE INFO

Keywords:

Non-*Saccharomyces* yeast
β-Glucosidase
β-Lyase
Biotransformation
Thiols
Beer aroma

ABSTRACT

Recently, non-*Saccharomyces* yeasts have been used in beer brewing to enhance the flavor complexity and highlight the unique flavor. *Wickerhamomyces anomalus* M3 with high β-glucosidase activity and *Pichia kluyveri* B21 with β-lyase activities, screened from fermented grains of *Baijiu*, were used in beer brewing process. Both strains could tolerate the beer brewing environment. Due to its β-glucosidase activity, M3 significantly increased terpenes levels in beer. In final beer, the levels of linalool, nerol and geraniol were increased to 170.44 μg/L, 3.77 μg/L and 4.70 μg/L, respectively. Furthermore, 7.75 μg/L of β-citronellol was newly generated, collectively enhancing the beer's floral aroma profile. The β-lyase activity of B21 facilitated production of volatile thiols including 4-mercapto-4-methylpentan-2-one (4.61 ng/L), 3-mercaptohexan-1-ol (270.52 ng/L), and 3-mercaptohexyl acetate (9.75 ng/L), contributing tropical fruit aromas of beer. Applying non-*Saccharomyces* yeasts in beer fermentation improved the conversion of hop/wort precursors into esters, terpenes, and volatile thiols, markedly improving floral and tropical fruit aromas and overall beer flavor quality.

1. Introduction

In recent years, the fermentation traits of non-*Saccharomyces* yeasts have garnered growing interest (Basso et al., 2016; Praia et al., 2022). Research indicates fermentation products of these yeasts would influence the levels of alcohols, esters, aldehydes, organic acids, and other volatile compounds, imparting more distinctive flavors of alcoholic beverages (Larroque et al., 2021; Pirrone et al., 2025). Moreover, compared to *Saccharomyces cerevisiae*, non-*Saccharomyces* yeasts are capable of secreting a variety of extracellular enzymes. These enzymes interact with aroma precursors in wort and hops, activating their aromatic properties and thereby shaping the overall aroma of fermented beverages (Belda et al., 2016). In beer brewing, yeast plays a crucial role in defining the final flavor (Herkenhoff et al., 2023) and quality of the beer. The history of beer fermentation has evolved from mixed-culture processes to dominant pure-culture methods, with *S. cerevisiae* (ale yeast) and *S. pastorianus* (lager yeast) being most widely employed today. However, with the increasing demand for diverse products, many different yeast strains are now being applied in beer brewing (Klimczak et al., 2024; Peces-Pérez et al., 2022).

During the beer fermentation process, yeast converts substances

from wort and hops into higher alcohols, esters, volatile glycosides, volatile thiols, and other compounds under enzymatic action, imparting unique aromatic characteristics to beer. In the biotransformation pathways, β-glucosidase and β-lyases play crucial roles. β-Glucosidase can hydrolyze substrates containing β-D-glycosidic bonds, releasing glucose and free glycoside ligands with aromatic properties and promoting the generation of aroma compounds (Han et al., 2023). Studies showed that β-glucosidase was commonly found in yeast, with non-*Saccharomyces cerevisiae* typically exhibiting higher enzymatic activity than *S. cerevisiae* (Karaulli et al., 2024; Xiaohong et al., 2020). β-Glucosidase produced by *Brettanomyces custersii* can hydrolyze glycosidic compounds from malt and hops, generating geraniol and β-citronellol (Serra Colomer et al., 2020). When acting on hops, whether in pure culture or co-culture with *Saccharomyces* yeasts, it results in the highest release levels of linalool, methyl salicylate, 1-octen-3-ol, and cis-3-hexen-1-ol. β-Lyases catalyze the cleavage of carbon-sulfur bonds linked to glutathione or cysteine, releasing volatile thiols from their precursors (Krogerus et al., 2023). During wine fermentation, *Metschnikowia pulcherrima*, *Torulaspora delbrueckii*, and *Lachancea thermotolerans* were reported to produce 3-mercaptohexanol or 4-methyl-4-mercaptopentan-2-one, contributing aromas of gooseberry, pineapple, grapefruit, passionfruit, and citrus to

* Corresponding author at: Laboratory of Brewing Science and Technology Jiangnan University, 1800 Lihu Avenue, Wuxi, Jiangsu 214122, China.

E-mail address: jjwang@jiangnan.edu.cn (J. Wang).

<https://doi.org/10.1016/j.foodres.2026.118443>

Received 27 October 2025; Received in revised form 5 January 2026; Accepted 14 January 2026

Available online 16 January 2026

0963-9969/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

the wine (Christiaens et al., 2025; Zott et al., 2011). In beer brewing, only 0.1–0.5% of bound thiols in wort are converted into free thiols, while 10–20% of volatile thiols exist in bound forms, requiring β -lyases to release these aroma compounds during fermentation (Gros et al., 2013). Low-alcohol and non-alcoholic beer attract much interests recently, and either traditional distilling or limited fermentation methods tend to cause the loss of characteristic aroma compounds such as esters and terpenes. Screening and development of functional brewing strains could address this issue (Costantini et al., 2025). The development of multi-omics techniques provided strong support for the precise screening of brewing strains (Herkenhoff et al., 2023).

Chinese traditional fermented grains harbor a rich diversity of microbial resources (Praia et al., 2022), among which yeasts serve as the primary microbial population responsible for aroma formation during the production of traditional *Baijiu*. In the fermentation grains of *Baijiu*, yeasts such as *Pichia*, *Candida*, *Kazachstania*, and *Wickerhamomyces* have been identified to play crucial roles in contributing to the complex flavor profile, including the formation of esters and aromatic compounds (Liang et al., 2025; Wang et al., 2025). This study focused on screening yeasts with high β -glucosidase and β -lyase activities from fermented grains of strong flavor *Baijiu* and evaluated the adaptability of the strain to beer brewing.

2. Materials and methods

2.1. Screen of yeast strains from fermented grains of Baijiu

Yeast strains were screened from fermented grains in *Baijiu* production due to its rich microbial diversity especially yeast resources. The samples were collected from a *Baijiu* Brewery in Wuxi, Jiangsu, China. Samples were enriched 28 °C for 48 h. Gradient dilutions with different concentrations were prepared, setting the dilution gradients from 10^{-1} to 10^{-7} . An appropriate amount of the dilution was spread on YPD plates containing ampicillin and incubated at 28 °C for 2 d. Based on morphological characteristics, strains exhibiting good growth and have typical yeast morphology, such as spherical, oval, or elliptical shapes, with milky-white colony color, and rough or smooth edges, were selected. These selected strains were inoculated into YPD liquid medium and incubated overnight at 28 °C.

Strains with higher β -glucosidase activity were roughly screened by inoculating 50 μ L of the seed culture into 1 mL of aesculin medium as described in Zheng et al. (Zheng et al., 2012) and incubated at 28 °C for 72 h. Darker color indicated higher β -glucosidase activity. Strains with higher β -lyase were screened by growing yeasts on Yeast Carbon Base-S-methyl-L-cysteine (YCB-SMC) agar plate for 72 h at 28 °C. Larger colonies indicate higher β -lyase activity (Belda et al., 2016).

The growth of yeasts was tested on YNB with different carbon sources including glucose, maltose, lactose, sucrose, fructose, and YNB without amino acids and ammonium sulfate with different nitrogen sources including peptone, ammonium sulfate, potassium nitrate, ammonium chloride, urea. Yeasts were cultured overnight in YPD medium at 28 °C and transferred to H₂O for 4 h. After gradient dilution process, the strains were inoculated on solid medium with different carbon/nitrogen sources and incubated at 28 °C for 2 d. Strains grown on YPD plate were used as control.

2.2. β -glucosidase and β -lyase activity assay

The β -glucosidase activity was measured as described by Decao et al. (Decao et al., 2018). Yeast strains were cultured in YPD medium for 72 h, and the fermentation broth was collected and centrifuged at 8000 r/min for 8 mins. The supernatant was collected and used as crude enzyme extraction. Mix 100 μ L of crude glucosidase enzyme solution, 375 μ L of citrate-phosphate buffer (pH 5.0), and 125 μ L of 1 mmol/L p-NPG and incubate the mixture in a water bath at 40 °C for 30 mins, 0.5 mL of 1 mol/L Na₂CO₃ is immediately added to stop the reaction. The

absorbance is measured at 400 nm using a microplate reader. The β -glucosidase activity is calculated based on a standard curve. One unit (U) of β -glucosidase activity is defined as the amount of enzyme required to catalyze the formation of 1 μ mol of p-nitrophenol per minute at 40 °C and pH 5.0.

β -lyase activity was calculated by measuring yeast growth in cysteine medium and cysteine consumption during the growth period as described by Krogerus et al. (Krogerus et al., 2023). The enzyme activity assay medium contains 0.17% YNB without (NH₄)₂SO₄ and amino acids, 1% glucose, 0.01% 5-phosphate pyridoxal, and 15 mmol/L cysteine which became the culture medium, then 1 mL of this medium and 100 μ L of bacterial suspension were added to a 96 well plate, sealed with BreathEasy membrane, and reacted at 25 °C for 5 days. The L-cysteine content of the culture medium was measured using the Ellman's reagent assay with 5,5'-dithiobis (2-nitrobenzoic acid) (DTNB). The unit of β -lyase activity (U) is defined as the amount of enzyme consumed to catalyze 1 μ mol of cysteine per 120 h at 25 °C.

2.3. Molecular identification of strains

Genomic DNA of yeast strains were extracted using MDT-MD01–01 Microbial Genome DNA Extraction Kit (column method) following manufacturer's instruction. The ITS sequence was amplified using PCR. The primers for PCR amplification were designed as follows: ITS1: 5'-TCCGTAGGTGAACCTGCGG-3' and ITS4: 5'-TCCTCCGCTTATTGATATGC-3'. The extracted DNA was directly used for PCR, and the reaction system was prepared as follows: 1 μ L ITS1, 1 μ L ITS4, 0.5 μ L genomic DNA, 10 μ L ddH₂O, and 12.5 μ L rTaq polymerase. The PCR cycling conditions were as follows: an initial denaturation at 94 °C for 5 min, followed by 35 cycles of denaturation at 94 °C for 45 s, annealing at 55 °C for 40 s, and extension at 72 °C for 60 s. A final extension step at 72 °C was carried out for 10 min. The PCR products were analyzed using 1.0% agarose gel electrophoresis. The amplified products were sent to Jinweizhi Biotechnology Co., Ltd. (Suzhou, Jiangsu, China) for sequencing.

2.4. Stress resistant assays

Temperature tolerance, ethanol tolerance and pH tolerance of the yeast strain were evaluated. A yeast culture of 3×10^8 CFU/mL was inoculated into YPD liquid medium at 1%(v/v) inoculation rate and cultured at different temperatures ranging from 4 to 37 °C. Ethanol tolerance was tested in YPD medium supplemented with ethanol ranging from 0% to 15%. pH tolerance was tested by culturing yeast in YPD medium adjusted to pH values ranging from pH 2–10.

2.5. Lab-scale beer fermentation test

Whole-malt wort was prepared from pale ale malt for beer fermentation test. Malt was mashed at 63 °C for 1 h and mashed out at 78 °C for 10 mins. The wort was transferred to a kettle and boiled for 1 h with 3 g/L hops added 5 mins before the end. The hop type is Cascade, with an alpha acid content of 5.6–8.8%. Dilute the malt extract with sterile water, adjust to 12°P, and cool to room temperature. Yeast was fermented in wort at 22 °C for 8 d. After fermentation, basic physical and chemical indicators such as fermentation degree, alcohol content, and free amino nitrogen(FAN) are measured. The above indicators were measured using Alcozyzer Beer Analyzing System. FAN concentration was determined using the ninhydrin method.

2.6. Determination of volatile compounds

Beer samples were filtered and degassed. An 8 mL sample was mixed with 2.0 g of NaCl in a 20 mL headspace vial, followed by the addition of 5 μ L of a 25 mg/L internal standard solution of 2-octanol. The mixture was thoroughly mixed and then analyzed by gas chromatography–mass

spectrometry (GC–MS) as described in our previous study (Guan et al., 2023). A TG-WAXMS A column (60 m × 0.25 mm × 0.25 μm) was used for gas chromatography analysis. The initial oven temperature was set to 40 °C for 1 min, then increased to 180 °C at a rate of 3 °C/min, followed by an increase to 230 °C at 20 °C/min, which was held for 15 mins. High-purity helium (purity >99.99%) was used as the carrier gas at a flow rate of 1.0 mL/min. The injector temperature was set to 250 °C, operated in splitless mode, with a solvent delay of 1 min. The mass spectrometer used an electron ionization (EI) source with the ion source temperature at 230 °C, the quadrupole temperature at 150 °C, and the interface temperature at 280 °C. The emission current was 34.6 μA, with electron energy set at 70 eV. The mass scan range was from 29 to 500 amu.

The total ion chromatogram (TIC) of beer flavor compounds was obtained through GC–MS analysis. Flavor compounds were identified by comparing the results with the National Institute of Standards and Technology (NIST) 2014 mass spectrometry library, combined with retention index (RI) data. The relative content of flavor compounds was determined by normalizing the peak areas of the compounds to the internal standard using the peak area normalization method.

2.7. Measurement of thiols in beer

Thiols compounds including 3-mercaptohexan-1-ol (3MH), 3-mercaptohexyl acetate (3MHA) and 4-mercapto-4-methylpentan-2-one (4MMP) were measured with a modified method as described by Liu et al. (Liu et al., 2024). The finished beer was centrifuge for 5 min at 600*g. Then 200 μL of 70 mmol/L EDTA-2Na solution and 200 μL of 10 mmol/L 4,4'-dipyridyl disulfide (DTDP) derivatization solution were added to the supernatant of the fermentation broth. The mixture was agitated at room temperature at 120 rpm for 3 h, filtered through filter paper, and purified with 500 mg Strata® phenyl SPE cartridge (55 μm, 3 mL, Phenomenex LTD, Aschaffenburg, Germany). The SPE cartridge was conditioned sequentially with 6 mL of methanol and 6 mL of water. After loading with the sample, the cartridge was washed with 3 mL of H₂O followed by 3 mL of 50% methanol aqueous solution. Finally, elution was performed with 1.5 mL of methanol, and the eluate was analyzed by LC-MS/MS. The LC-MS/MS system comprised an Agilent 1290 Infinity II high-performance liquid chromatography system with a C18 column and an Agilent 6470 three-four pole mass spectrometer. The mobile phase consisted of 0.1% formic acid in water (Phase A) and methanol (Phase B), with gradient elution starting at 85% Phase A and increasing to 100% Phase B over 12 mins at a flow rate of 0.3 mL/min. Mass spectrometry was conducted under the following conditions: gas temperature of 280 °C, gas flow of 10 L/min, nebulizer pressure of 0.3 MPa, sheath gas heater temperature of 350 °C, sheath gas flow of 12 L/min, and a capillary voltage of 2500 V. Multiple reaction monitoring (MRM) with electrospray ionization (ESI) was used for compound detection.

2.8. Data analysis

Data were presented as mean, ±standard deviations (SD) and the means were calculated from at least three independent assays. To determine any significant differences between treatments, data was analyzed by one-way ANOVA with post hoc testing using SPSS 26.0. Statistically significant differences were reported at *P* < 0.05. The semi-quantitative heat map of cluster analysis was generated using TTools software. OAV = concentration of substance/aroma threshold of the substance.

3. Result

3.1. Screening of yeast strains with β-glucosidase and β-lyase activities

In this study, 90 strains were isolated from fermented grains in Baijiu. Based on morphological observations, 48 strains were selected. As shown in Table 1, Strains M3, 4, and 10 exhibited significantly higher

Table 1

The β-glucosidase and β-lyase activities of screened strains.

Strain	β-glucosidase activity (U/g)	β-lyase activity (U/g)	Strain	β-glucosidase activity (U/g)	β-lyase activity (U/g)
M1	5.36 ± 0.12 ^{defg}	16.58 ± 0.04 ^q	P11	4.26 ± 0.85 ⁱ	75.46 ± 0.1 ^k
M2	5.71 ± 0.09 ^{cd}	16.77 ± 0.13 ^q	P12	3.27 ± 0.13 ^{mn}	84.75 ± 0.12 ^h
M3	7.01 ± 0.14 ^a	21.4 ± 0.03 ⁿ	P13	3.06 ± 0.76 ^{no}	112.29 ± 0.31 ^c
M4	6.18 ± 0.20 ^b	18.69 ± 0.06 ^p	P14	2.32 ± 0.14 ^{qr}	71.29 ± 0.24 ⁱ
M5	5.54 ± 0.22 ^{def}	13.99 ± 0.3 ^{tuv}	P15	3.15 ± 0.34 ⁿ	79.58 ± 0.14 ^j
M6	6.02 ± 0.23 ^b	14.15 ± 0.1 ^{stuv}	ND1	2.75 ± 0.20 ^{op}	10.83 ± 0.17 ^w
M7	5.34 ± 0.24 ^{efg}	14.61 ± 0.2 ^{rst}	ND2	2.60 ± 0.31 ^{pq}	11.17 ± 0.04 ^w
JT1	4.34 ± 0.22 ^{ij}	14.59 ± 0.13 ^{rst}	ND3	3.24 ± 0.22 ^{mn}	10.10 ± 0.06 ^x
JT2	5.19 ± 0.21 ^{fg}	14.34 ± 0.21 ^{rstuv}	ND5	3.76 ± 0.17 ^{kl}	6.67 ± 0.13 ^y
JT3	5.34 ± 0.19 ^{efg}	13.79 ± 0.14 ^{uv}	ND6	4.36 ± 0.15 ⁱ	10.97 ± 0.11 ^w
JT4	5.15 ± 0.18 ^{fg}	14.23 ± 0.08 ^{stuv}	B21	5.22 ± 0.34 ^{efg}	157.92 ± 0.13 ^a
JT5	5.36 ± 0.18 ^{defg}	14.36 ± 0.04 ^{rstuv}	B22	3.57 ± 0.48 ^{lm}	112.23 ± 0.02 ^c
JT6	5.58 ± 0.19 ^{de}	14.17 ± 0.10 ^{stuv}	B23	4.27 ± 0.12 ^{ij}	119.07 ± 0.09 ^b
JT7	1.20 ± 0.15 ^{wxy}	14.75 ± 0.13 ^{rs}	B24	3.27 ± 0.85 ^{mn}	104.77 ± 0.13 ^d
P1	1.32 ± 0.11 ^{vw}	99.79 ± 0.17 ^e	B25	1.65 ± 0.23 ^{uv}	70.46 ± 0.13 ^m
P2	3.11 ± 0.28 ⁿ	86.78 ± 0.09 ^g	4	5.96 ± 0.18 ^{bc}	20.44 ± 0.16 ^o
P3	2.07 ± 0.18 ^{rs}	84.47 ± 0.06 ^h	10	6.11 ± 0.16 ^b	20.82 ± 0.2 ^{no}
P4	2.01 ± 0.15 ^{rst}	80.61 ± 0.11 ⁱ	JY1	4.72 ± 0.22 ^h	14.97 ± 0.12 ^r
P5	5.36 ± 0.18 ^{defg}	95.12 ± 0.10 ^f	JY2	5.04 ± 0.17 ^{gh}	13.95 ± 0.24 ^{tuv}
P6	2.11 ± 0.24 ^{rs}	84.73 ± 0.03 ^h	JY3	5.04 ± 0.17 ^{vwx}	16.36 ± 0.14 ^q
P7	1.21 ± 0.52 ^{wxy}	104.56 ± 0.21 ^d	JY4	0.92 ± 0.09 ^y	14.42 ± 0.16 ^{rstu}
P8	0.97 ± 0.03 ^{xy}	94.94 ± 0.20 ^f	JY5	1.84 ± 0.22 ^{stu}	13.65 ± 0.12 ^v
P9	1.38 ± 0.51 ^{vw}	86.98 ± 0.07 ^g	S189	3.41 ± 0.11 ^{jk}	10.87 ± 0.16 ^w
P10	1.74 ± 0.12 ^{tu}	70.41 ± 0.02 ^m	US05	3.83 ± 0.26 ^{kl}	13.98 ± 0.15 ^{tuv}

Values represent the average ± standard deviation (*n* = 3). Analyses of variance (ANOVA) were performed independently for each of the fermentations. The different letters in the ANOVA indicate significant differences for the set of samples.

β-glucosidase activities compared to the other strains. Furthermore, the β-lyase activities of strains B21, B23, P13 were notably higher than those of the remaining strains. Based on these findings, strains M3, 4, 10 and strain B21, B23, P13 were selected for further study due to their enhanced enzymatic activities. PCR was used to amplify the ITS region of these three strains and sent out for sequencing (Herkenhoff et al., 2023). As shown in Fig. 1, strain M3 was defined as *W. anomalus*, Strain 4 was defined as *Hanseniaspora uvarum*, strain 10 was defined as *H. opuntiae*, strain B21 was defined as *P. kluyveri*, strain B23 was defined as *P. kluyveri*, and strain P13 was defined as *P. kudriavzevii*.

3.2. Effect of non-Saccharomyces yeasts on the formation of flavor substances

Since the screened non-Saccharomyces yeasts showed relatively high β-glucosidase and β-lyase activity, they were tested for transformation of

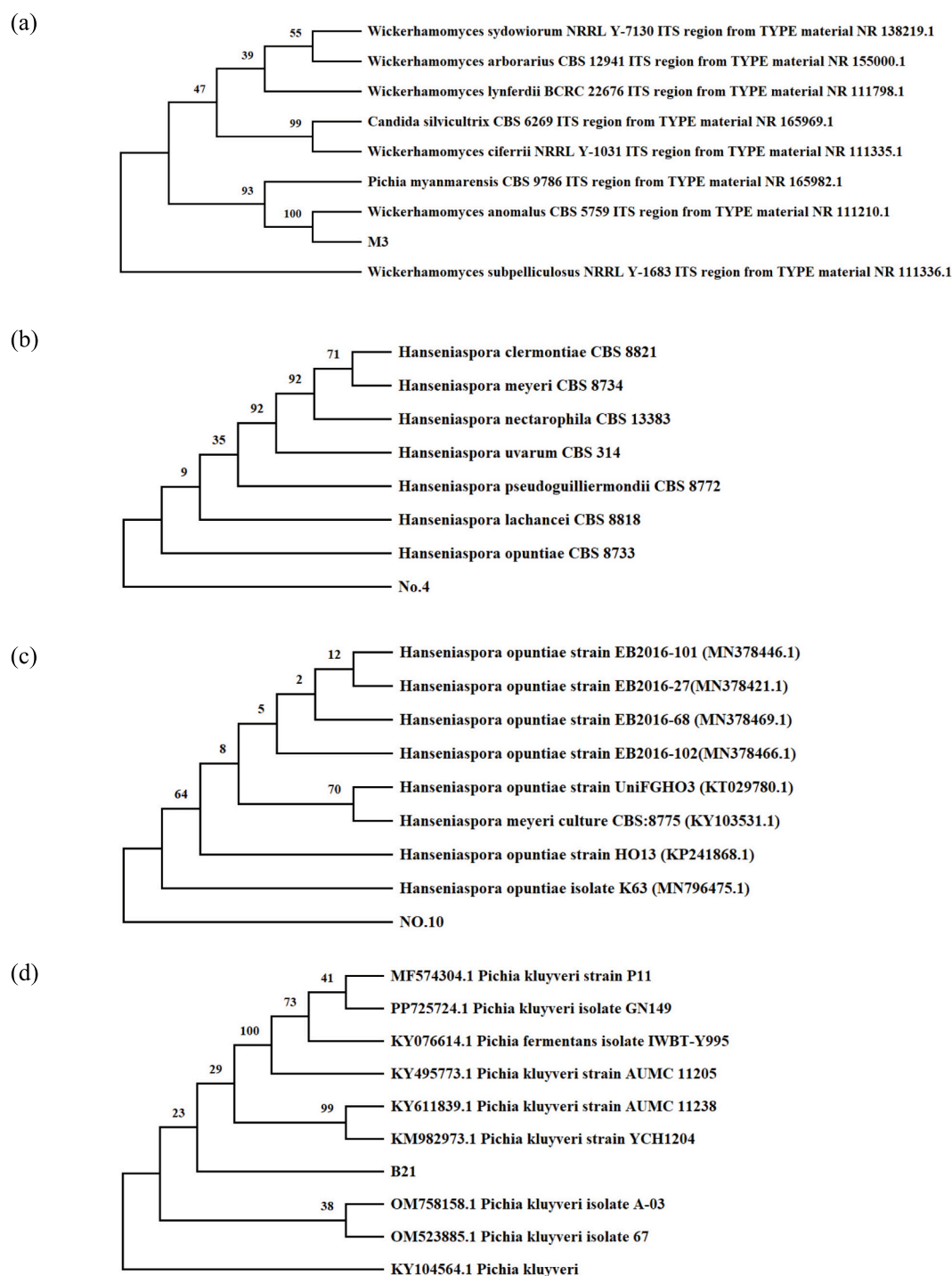


Fig. 1. Phylogenetic tree of non-*Saccharomyces cerevisiae* strain M3(a), 4(b), 10(c), B21(d), B23(e) and P13(f).

hop compounds in hop water. Commercial strains S189, US05 and LA01 were used as controls. As shown in Fig. 2 and Table 2, the concentrations of 14 compounds such as octanoic acid, dihydro- β -ionone, t-cadinol, (–)-perillyl alcohol, caryophyllene oxide, cedarwood oil, α -humulene and others showed no significant differences between the hop matrix and the fermented samples, suggesting these compounds stayed stable from hops to beer brewed in this experiment. In contrast, 11 ester compounds were newly generated through yeast biotransformation, which were ethyl acetate, isobutyl acetate, isobutyl isobutyrate, isopentyl acetate, butyl isobutyrate, isoamyl isobutyrate, isoamyl n-

butyrate and others. Additionally, 6 terpene compounds including linalool, geraniol, nerol, α -terpineol, β -citronellol, β -phellandrene, and three volatile thiols including 4MMP, 3MH, and 3MHA were identified (Herkenhoff et al., 2024).

Compared with the hop matrix, yeast biotransformation differentially increased the yield and diversity of volatile compounds. Regarding the number of volatile compounds, M3 and B21 (20 compounds) were the most among all the strains. Among all, M3 yielded the highest concentrations of ethyl acetate (28.46 mg/L), isoamyl acetate (286.63 μ g/L), linalool (2.59 mg/L), and geraniol (28.35 μ g/L). The linalool,

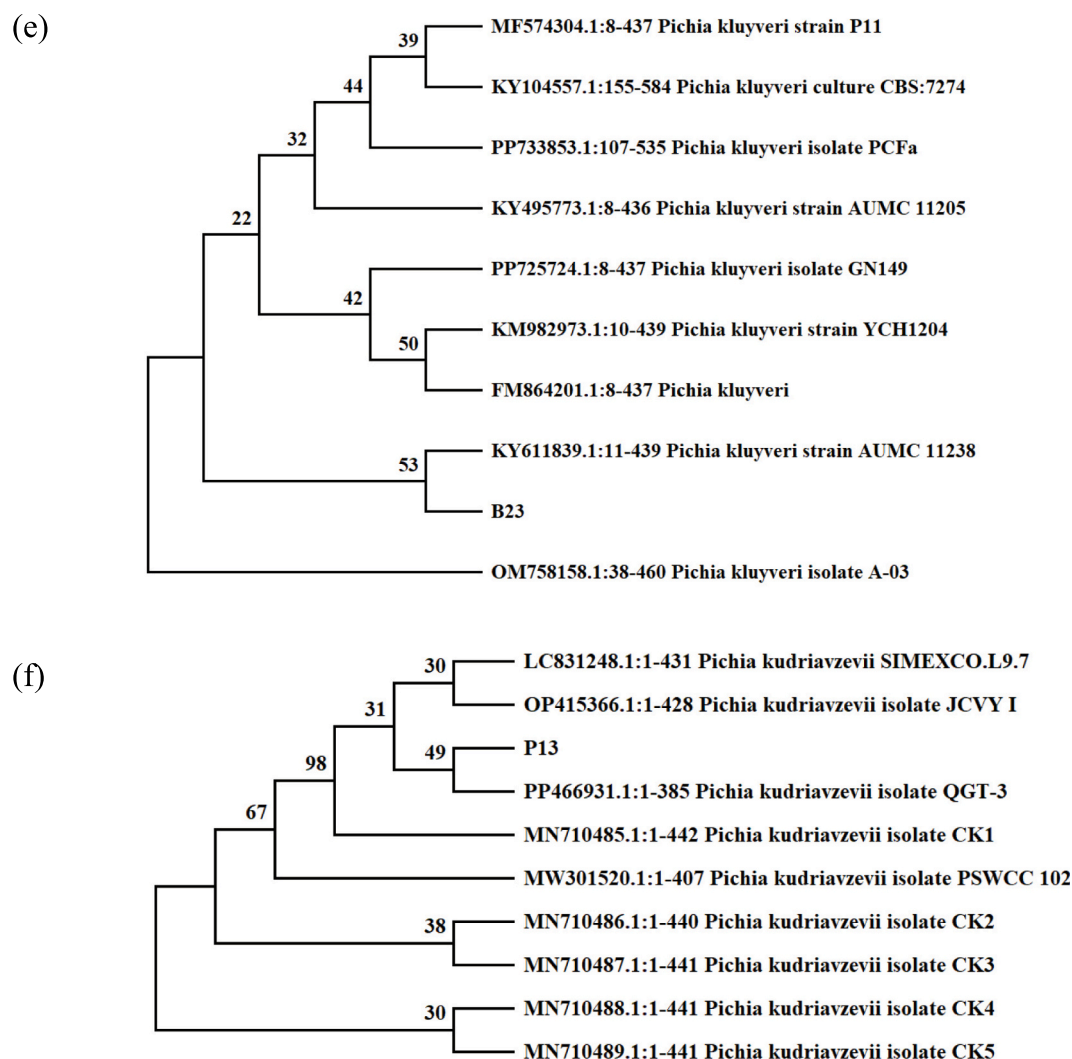


Fig. 1. (continued).

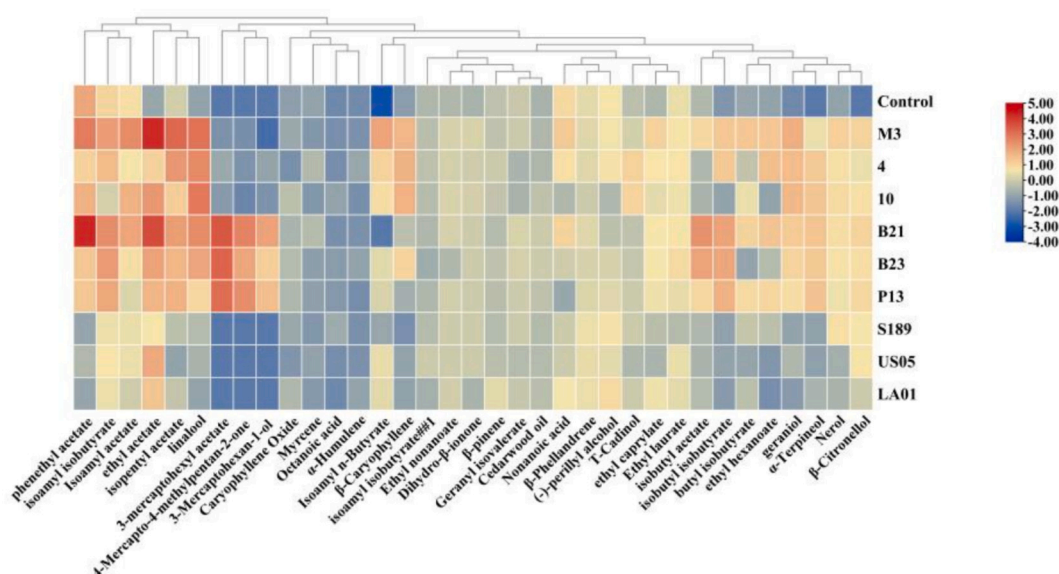


Fig. 2. Semi-quantitative cluster analysis of volatile compounds in hop water cultured with different strains.

nerol and geraniol concentration increased by 1.32 mg/L, 0.79 μ g/L and 12.08 μ g/L, respectively. Furthermore, β -citronellol, not originally

Table 2

Content of flavor compounds produced by different strains in hop matrix.

Compounds	Control	M3	4	10	B21	B23	P13	S189	US05	LA01
Ethyl acetate (μg/L)	0	28,455.09 ± 6.88 ^a	16,291.4 ± 3.32 ^a	18,274.74 ± 5.78 ^a	20,011.23 ± 6.47 ^a	18,212.23 ± 3.53 ^a	14,256.21 ± 6.36 ^a	8434.57 ± 4.67 ^a	17,226.47 ± 3.67 ^a	17,287.6 ± 5.73 ^a
Isopentyl acetate (μg/L)	0	286.63 ± 3.44 ^c	278.77 ± 7.27 ^c	265.43 ± 3.38 ^c	280.34 ± 3.68 ^c	279.57 ± 2.37 ^c	200.58 ± 5.89 ^f	264.01 ± 1.54 ^c	267.3 ± 1.35 ^c	13.1 ± 0.35 ^f
Linalool (μg/L)	1267.76 ± 2.55 ^a	2582.89 ± 2.57 ^b	1182.12 ± 9.30 ^b	1293.33 ± 2.86 ^b	1172.52 ± 5.26 ^b	1046.2 ± 3.71 ^b	790.46 ± 6.42 ^b	576.751 ± 2.56 ^b	463.9 ± 1.56 ^b	304.76 ± 1.86 ^b
Nerol (μg/L)	0.64 ± 0.24 ^d	1.43 ± 0.23 ^h	0.77 ± 0.02 ^j	1.28 ± 0.02 ^h	1.11 ± 0.01 ^j	1.02 ± 0.02 ^j	0.52 ± 0.01 ^j	0.75 ± 0.01 ^h	0.35 ± 0.02 ⁱ	0.4 ± 0.02 ^h
Geraniol (μg/L)	16.27 ± 1.45 ^b	28.35 ± 0.25 ^g	15.31 ± 0.21 ⁱ	18.3 ± 0.02 ^g	15.22 ± 0.02 ^g	15.02 ± 0.04 ^g	15.24 ± 0.01 ^g	15.06 ± 1.24 ^f	17.73 ± 0.68 ^f	14.35 ± 0.26 ^e
β-Citronellol (μg/L)	0	244.83 ± 1.34 ^d	230.42 ± 1.23 ^d	239.28 ± 1.02 ^d	250.56 ± 2.65 ^e	231.21 ± 2.76 ^e	249.1 ± 2.86 ^d	100.6 ± 1.35 ^d	103.11 ± 1.46 ^d	140.35 ± 1.46 ^c
α-Terpineol (μg/L)	0	100.67 ± 3.34 ^e	168.43 ± 1.64 ^e	180.39 ± 2.44 ^e	215.57 ± 2.51 ^f	170.35 ± 1.78 ^f	216.57 ± 1.57 ^e	80.7 ± 0.15 ^e	77.25 ± 2.67 ^e	80.7 ± 1.46 ^d
4-mercapto-4-methylpentan-2-one (ng/L)	0.2 ± 0.01 ^f	0.1 ± 0.01 ⁱ	0.2 ± 0.02 ^l	0.1 ± 0.01 ^j	4.2 ± 0.32 ⁱ	3.4 ± 0.29 ⁱ	4.01 ± 1.02 ⁱ	0.08 ± 0.01 ^j	0.2 ± 0.01 ^j	0.2 ± 0.01 ^j
3-mercaptohexan-1-ol (ng/L)	2.6 ± 0.04 ^c	60.2 ± 1.34 ^f	90.2 ± 1.54 ^f	77.2 ± 2.56 ^f	264.3 ± 1.89 ^d	260 ± 1.29 ^d	255 ± 1.20 ^c	7.02 ± 0.02 ^g	9.02 ± 0.02 ^g	12.01 ± 0.01 ^j
3-mercaptohexyl acetate (ng/L)	0.31 ± 0.01 ^e	0.1 ± 0.01 ⁱ	0.5 ± 0.02 ^k	0.3 ± 0.03 ⁱ	9.20 ± 0.81 ^h	8.02 ± 0.86 ^h	6.01 ± 0.86 ^h	0.3 ± 0.01 ⁱ	0.4 ± 0.01 ^h	0.31 ± 0.01 ⁱ

Values represent the average ± standard deviation ($n = 3$). The different letters in the ANOVA indicate significant differences for the set of samples.

present in the hop matrix, was detected at concentrations of 244.83 μg/L (M3). This aligns with findings by K. Takoi et al. (Takoi et al., 2009) that β-citronellol is nearly absent in hops and wort but gradually increases during fermentation due to the conversion of geraniol to β-citronellol by yeast.

Yeast produces β-lyase, which catalyzes the production of volatile

thiols from cysteine precursors in hops. The majority of volatile thiols in beer come from hops, in both free and bound states. The difference between the two is that free thiols are directly released into beer through dry feeding to present aroma, while bound thiols require β-lyase to release these thiols in order to affect the flavor and aroma. Results showed that B21, B23, and P13 had high β-lyase activity. Highest

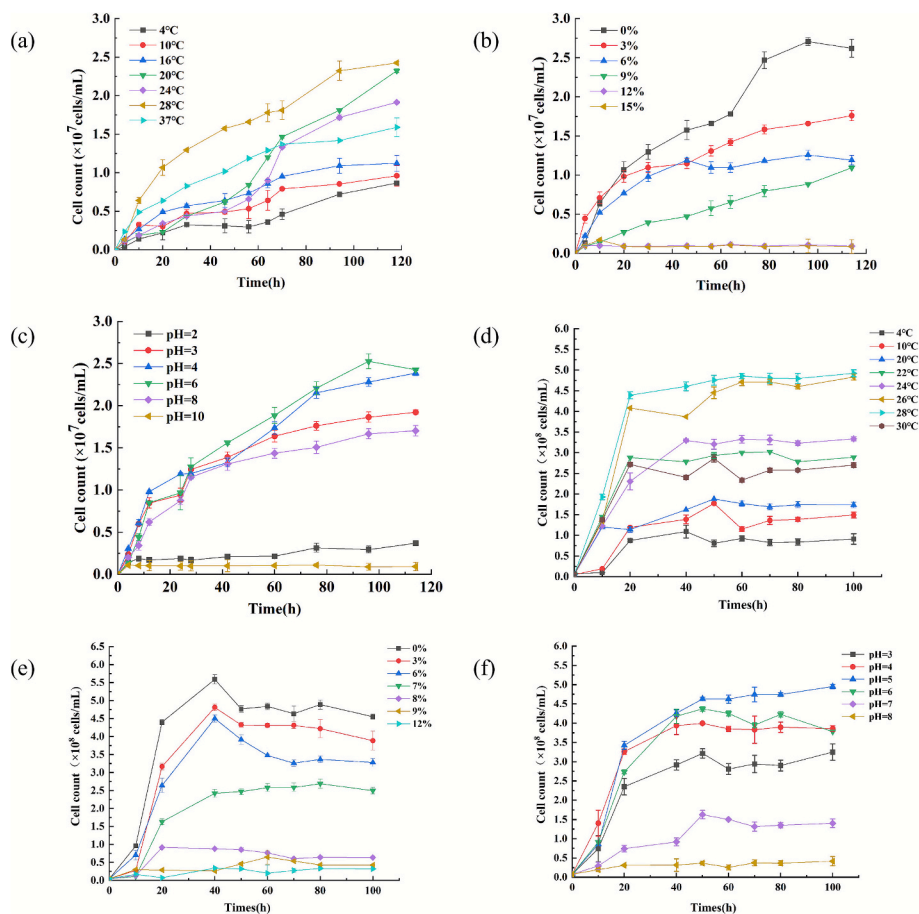


Fig. 3. Analysis of temperature tolerance (a), ethanol tolerance (b), and pH tolerance (c) of strain M3. Analysis of temperature tolerance (d), ethanol tolerance (e) and pH tolerance (f) of strain B21.

production of thiols were found in B21 matrix, where the concentrations of 4MMP, 3MH, and 3MHA were 4.2 ng/L, 264.3 ng/L, and 9.2 ng/L, respectively. These compounds exhibit aromas of blackcurrant, grapefruit, and passion fruit. Allen et al. (Allen et al., 2011) found that the content of volatile thiols varies significantly depending on the type of yeast used, with non-*Saccharomyces* generally producing higher levels of volatile thiols than *Saccharomyces* yeasts. α -Terpineol, which was not detected in the hops matrix, was detected in all fermentation samples, with the highest content of α -terpineol produced by B21 being 215.57 μ g/L. The release of monoterpenols could result from high temperature boiling or hydrolysis catalyzed by acids or enzymes. In beer brewing, geraniol is converted to β -citronellol and linalool, while linalool and nerol are isomerized to α -terpineol. A 250.56 μ g/L of β -citronellol were measured in B21 matrix, however, the linalool concentration was relatively low in B21 matrix, at 1.17 mg/L.

Strain M3 (*W. anomalus*, CGMCC No. 28666) and strain B21 (*P. kluyveri*, CGMCC No. 34028) were selected for further study.

3.3. Tolerance analysis of selected yeasts

Yeasts were tested under different stress conditions for their optimal growth conditions. Temperature is a crucial environmental factor affecting the growth and distribution of organisms. As shown in Fig. 3a and Fig. 3d, M3 grew well in the temperature range of 20–28 °C and B21 could grow in the range of 22–28 °C.

The alcohol toxicity in the fermentation broth is an important cause of the decline in cell viability. As shown in Fig. 3b and Fig. 3e, strain M3 could not grow when the ethanol concentration exceeds 12% (v/v), and strain B21 could not grow when the ethanol concentration exceeds 8% (v/v). pH affects the growth and fermentation capability of yeasts and as shown in Fig. 3c and Fig. 3f, both M3 and B21 could grow under pH 4–6. Above results indicate both M3 and B21 could be used for beer brewing.

In addition, both strains were tested for consumption of different carbon and nitrogen sources. As shown in Table 3, B21 and M3 can metabolize and utilize multiple carbon and nitrogen sources for growth, but there were significant differences in utilization efficiency. In terms of carbon source utilization ability, M3 and B21 has the best glucose utilization ability. In terms of nitrogen source utilization ability, M3 and B21 have the best utilization ability for peptone. Through this experiment, the growth characteristics of the strain can be understood, providing a theoretical basis for subsequent experiments.

3.4. Application of non-*Saccharomyces* yeast in low alcohol beer fermentation

The initial inoculation amount, fermentation temperature, and initial pH affect beer fermentation and the production of flavor compounds. These parameters were optimized for the application of the

Table 3
Carbon and nitrogen source utilization abilities of B21 and M3.

Medium composition	M3	B21
No carbon source	—	—
Glucose	+++	+++
Fructose	++	++
Maltose	++	+
Sucrose	+	+
No nitrogen source	+	+
Peptone	+++	+++
Ammonium chloride	++	+
Ammonium sulfate	++	+
Potassium nitrate	+	+
Urea	+	—

Evaluate the utilization of carbon and nitrogen sources by observing the growth of bacterial colonies on the agar plate. “+++” indicates strong utilization ability, “++” indicates strong utilization ability, “+” indicates average utilization ability, and “—” indicates inability to utilize.

strains in beer brewing.

3.4.1. Application of M3 in low alcoholic beer brewing

As mentioned above, the M3 strain could generate relatively higher amounts of esters. Therefore, esters' concentration was used for the optimization process for M3 fermentation. The initial pH was set at pH 3.5, 3.8, 4, 4.2, 4.5, and 5.5, relatively. The decrease in pH value led to a reduction in biomass, alcohol content, and fermentation degree, well as a decrease in ethyl acetate production. As shown in Fig. 4, when the initial pH value was below 5.5, yeast growth and fermentation were inhibited. Under pH 5.5 conditions, yeast achieved the highest biomass, CO₂ weight loss, fermentation degree, and alcohol content. When pH was 5.5, the highest content of ethyl acetate was 162.67 mg/L. pH value of 5.5 was chosen for fermentation.

Regarding the fermentation temperatures, 18 °C, 20 °C, 22 °C, 24 °C, and 26 °C were selected for brewing. As shown in Fig. 4, the highest amount of esters was 217.27 mg/L at 26 °C, however, high-temperature fermentation results in a poor overall flavor balance of the beer. Here 24 °C was chosen for further optimization.

As the inoculation amount increased, the ethyl acetate content showed a trend of first increasing and then decreasing. At an initial inoculation level of 7×10^6 CFU/mL, the ethyl acetate concentration peaked at 440.67 mg/L. Research showed that the sensory of beer will be better when the content of ethyl acetate is between 20.0 and 25.0 mg/L. Excess ethyl acetate content will leave beer with an unpleasant solvent smell. Meanwhile, the increases in alcohol content and fermentation efficiency were not significant with higher inoculation levels. Considering the economic efficiency of yeast usage, 5×10^6 CFU/mL was selected as the optimal inoculation amount. Therefore, by optimizing the fermentation conditions, it was found that under the conditions of pH 5.5, fermentation temperature of 24 °C, and inoculation volume of 5×10^6 CFU/mL, the yield of ethyl acetate was the highest, reaching 217.49 mg/L.

As shown in Table 4, the concentration of linalool in beer fermented with M3 increased by 7.15% compared to that fermented with US05. The concentration of β -citronellol in beer was 7.75 μ g/L. No β -citronellol was detected in beer fermented with the US05, indicating that the increased β -citronellol was likely entirely produced by the M3. The concentration of 3-mercaptohexanol in M3 fermented beer was 92.6 ng/L, which was 9.43% higher than that in US05 fermented beer.

Compared with LA01 beer, the concentrations of ethyl laurate, isoamyl laurate, and ethyl hexanoate were 157.67 μ g/L, 248.12 μ g/L, and 27.72 μ g/L, respectively. These substances were produced by strain M3 and were not detected in LA01 beer. The concentration of phenethyl alcohol in beer by strain M3 increased by 72.77% compared with strain LA01. In addition, the ethyl acetate content in the strain M3 beer increased significantly to 22.66 mg/L.

All beers were used for sensory evaluation with the non-alcoholic yeast LA01 and ale yeast US05 used as controls. As shown in Fig. 6, compared with the commercial yeasts, the beer fermented by M3 has an obvious floral aroma (Edgar Herkenhoff et al., 2024), which was related to the abundance of esters and terpenoids.

3.4.2. Application of B21 in low alcoholic beer brewing

In the fermentation using the B21 strain, the activity of β -lyase was used as a criterion for fermentation optimization. By setting different initial pH conditions for fermentation, the activity of β -lyase reached its peak on the first day and then stabilizes. As shown in Fig. 5, the enzyme activity increases with the increase of pH value, with lower levels at pH 4.0, 4.5, and 6.0, and higher activity at pH 5.0 and 5.5. The highest β -lyase activity, 144.01 U/g, was achieved at an initial pH of 5.5, which was consequently chosen as the optimal pH. The fermentation temperatures compared were 18 °C, 20 °C, 22 °C, and 24 °C. As the temperature increases, the activity of β -lyase correspondingly increases, reaching a maximum value of 153.15 U/g at 22 °C. Further increase in temperature to 24 °C resulted in a slight decrease in activity, indicating inhibition of

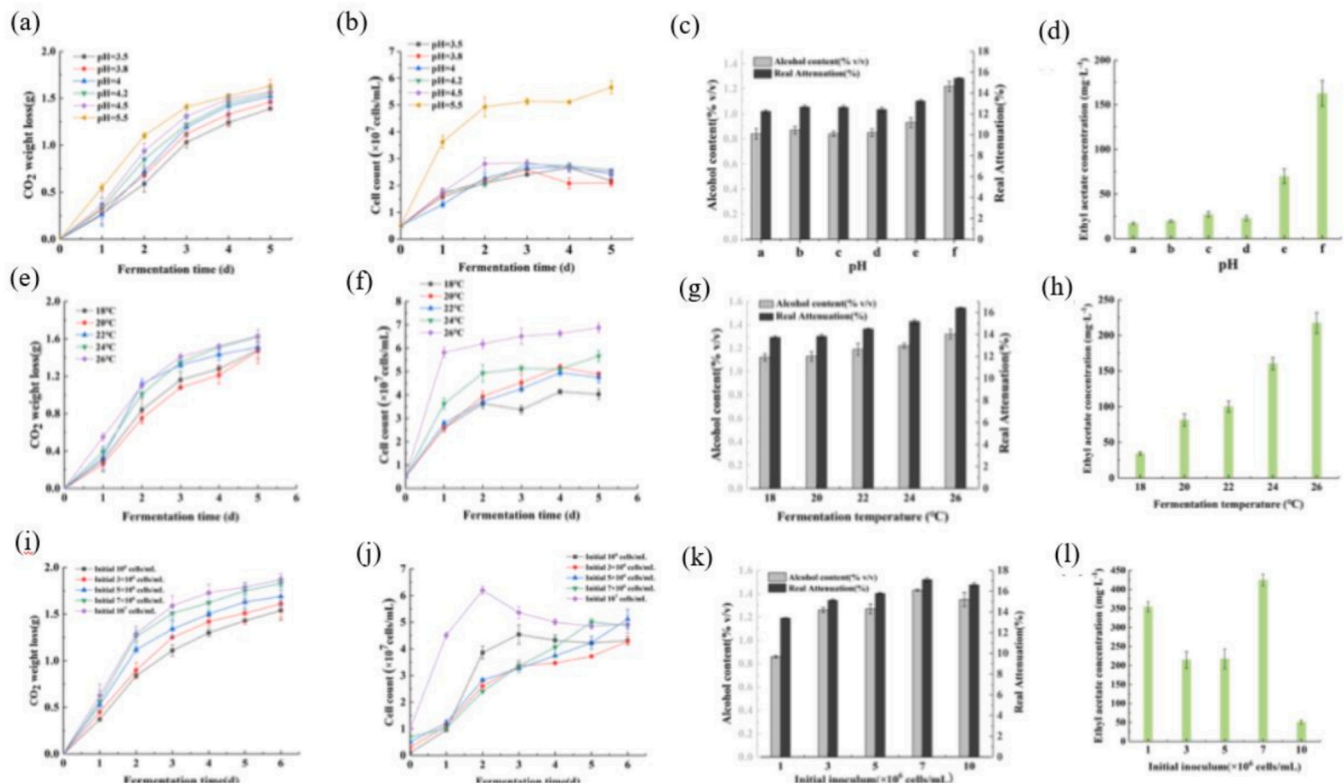


Fig. 4. Optimization of pH (a-d), fermentation temperature (e-h), and inoculation volume (i-l) of M3 in beer brewing.

Table 4
Analysis of volatile compounds in beer.

Compounds	M3 beer	US05 beer	B21 beer	LA01 beer	OAV (M3 beer)	OAV (US05 beer)	OAV (B21 beer)	OAV (LA01 beer)
Ethyl acetate (μg/L)	22,664.73 ± 21.63 ^a	11,719.68 ± 13.42 ^a	20,909.11 ± 31.23 ^a	10,040.25 ± 1.34 ^a	3.00	1.56	2.78	1.33
Isobutyl acetate (μg/L)	–	38.94 ± 1.78 ^l	44.71 ± 0.55 ^k	10.33 ± 0.24 ^l	–	0.02	0.03	0.01
Isoamyl acetate (μg/L)	281.97 ± 8.66 ^f	178.13 ± 3.57 ^f	786.96 ± 5.23 ^d	200.11 ± 1.34 ^f	9.29	5.93	26.23	6.67
Ethyl caprylate (μg/L)	9.78 ± 0.77 ^q	3.84 ± 0.54 ^p	–	1.34 ± 0.04 ⁿ	0.04	0.02	–	0.01
Ethyl caprate (μg/L)	6.43 ± 0.34 ^s	24.56 ± 21.42 ^m	230.02 ± 4.67 ^f	30.33 ± 1.23 ⁱ	0.03	0.12	1.15	0.15
Phenylethyl acetate (μg/L)	2.39 ± 0.07 ^v	1.75 ± 0.46 ^q	225.06 ± 2.55 ^g	–	0.01	0.01	0.9	–
Ethyl laurate (μg/L)	157.67 ± 8.79 ^j	103.21 ± 13.6 ^e	–	–	0.04	0.03	–	–
Isoamyl laurate (μg/L)	248.12 ± 6.36 ^h	–	–	–	1.24	–	–	–
Ethyl hexanoate (μg/L)	27.72 ± 4.25 ⁿ	19.46 ± 6.73 ⁿ	26.19 ± 0.16 ^l	–	0.14	0.09	0.13	–
Ethyl pelanoate (μg/L)	88.91 ± 3.38 ^l	89.58 ± 6.75 ⁱ	–	22.33 ± 1.23 ^j	0.07	0.07	–	0.02
Ethyl myristate (μg/L)	46.41 ± 2.16 ^m	57.81 ± 3.21 ^k	–	–	0.05	0.06	–	–
1-Propanol (μg/L)	5343.06 ± 23.56 ^c	6642.48 ± 16.38 ^b	–	4661.31 ± 12.33 ^c	0.2	0.3	–	0.18
2-Methyl-1-propanol (μg/L)	3905.49 ± 15.09 ^d	3610.25 ± 24.56 ^d	4563.24 ± 13.45 ^c	3735.12 ± 21.33 ^d	0.07	0.06	0.08	0.06
3-Methyl-1-butanol (μg/L)	2369.32 ± 31.54 ^e	2298.18 ± 15.66 ^e	–	2241.12 ± 12.31 ^e	0.04	0.04	–	0.04
Linalool (μg/L)	170.44 ± 9.87 ⁱ	100.07 ± 12.43 ^h	55.89 ± 1.56 ^j	75.24 ± 1.23 ^g	6.82	4.00	2.24	3.01
Nerol (μg/L)	3.77 ± 0.68 ^u	–	–	–	0.01	–	–	–
Geraniol (μg/L)	4.70 ± 0.44 ^t	–	–	–	0.1	–	–	–
Phenethyl alcohol (μg/L)	8737.01 ± 28.78 ^b	6091.23 ± 31.63 ^c	4967.43 ± 4.24 ^b	5057.21 ± 24.5 ^b	1.74	1.2	0.98	1.01
α-Terpinol (μg/L)	18.22 ± 2.36 ^o	–	95.26 ± 1.46 ⁱ	–	0.9	–	4.76	–
β-Citronellol (μg/L)	7.75 ± 1.58 ^r	–	–	–	0.96	–	–	–
1-Nonanol (μg/L)	–	–	3.71 ± 0.24 ^o	2.24 ± 1.34 ^m	–	–	0.41	0.25
2,4-Ditertbutyl phenol (μg/L)	13.43 ± 2.14 ^p	7.36 ± 1.45 ^o	–	20.24 ± 0.13 ^k	0.09	0.05	–	0.13
4-Mercapto-4-methylpentan-2-one (ng/L)	–	–	4.61 ± 0.42 ⁿ	–	–	–	3.07	–
3-Mercaptohexan-1-ol (ng/L)	92.60 ± 4.85 ^k	84.62 ± 6.93 ^j	270.52 ± 9.89 ^e	55.23 ± 0.13 ^h	1.68	1.53	4.92	1.00
3-mercaptohexyl acetate (ng/L)	–	–	9.75 ± 0.8 ^m	–	–	–	1.95	–
Free Amino Nitrogen (mg/L)	221.60 ± 1.22 ^g	–	190.00 ± 1.15 ^h	–	–	–	–	–

Values represent the average ± standard deviation ($n = 3$). Analyses of variance (ANOVA) were performed independently for each of the fermentations. The different letters in the ANOVA indicate significant differences for the set of samples.

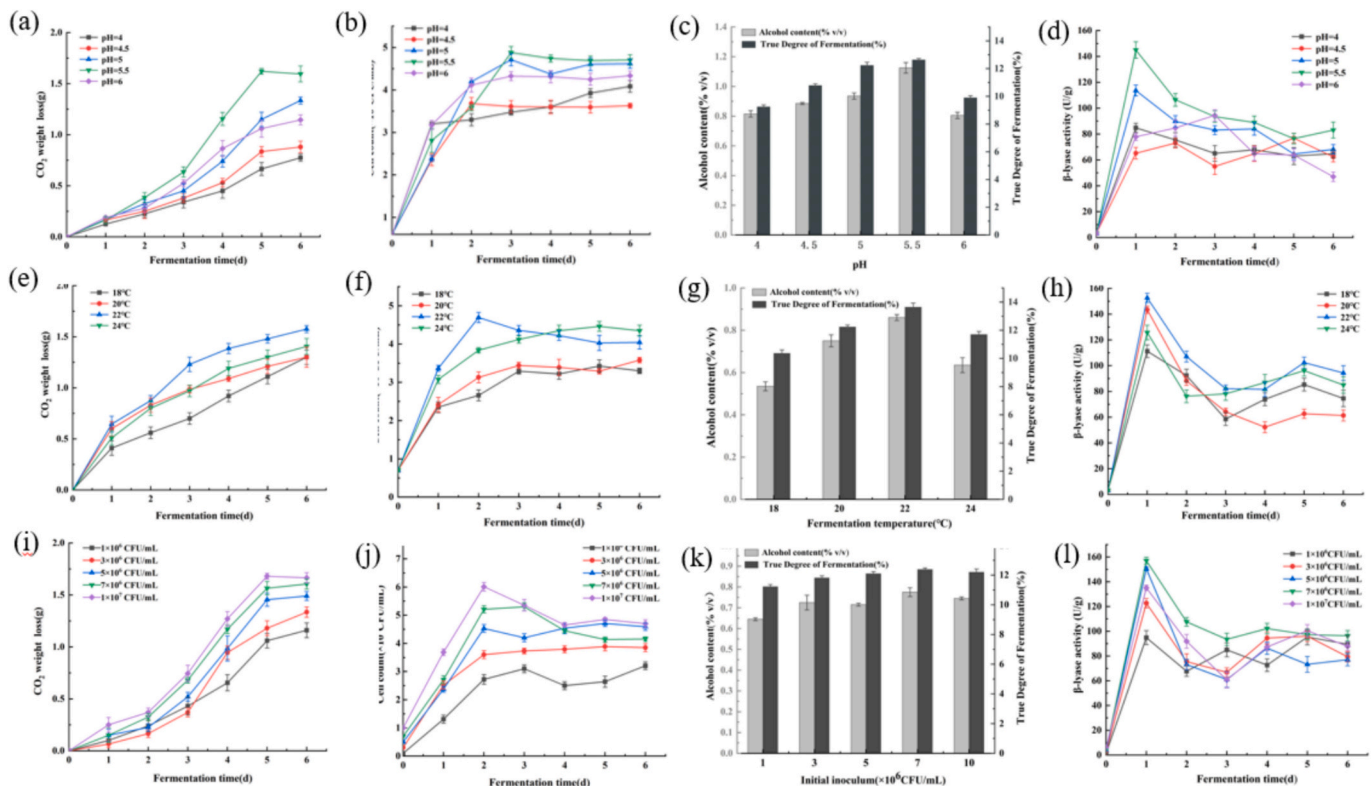


Fig. 5. Optimization of pH (a-d), fermentation temperature (e-h), and inoculation volume (i-l) of B21 in beer brewing.

yeast function at high temperatures. Therefore, 22 °C was determined as the optimal fermentation temperature.

Different inoculation amounts were compared with these being 1, 3, 5, 7, and 10 × 10⁶ CFU/mL for fermentation. The highest enzyme activity was observed with an inoculation amount of 7 × 10⁶ CFU/mL. In contrast, a lower inoculation rate of 1 × 10⁶ CFU/mL resulted in consistently low β-lyase activity due to limited yeast population and reduced enzyme production. Thus, 7 × 10⁶ CFU/mL was selected as the optimal inoculation amount, and the enzyme activity was 157.07 U/g. The optimal process for B21 fermentation consisted of an inoculation amount of 7 × 10⁶ CFU/mL, fermentation temperature of 22 °C, and initial pH of 5.5. This resulted in a maximum β-lyase activity of 157.07 U/g, alcohol content of 1.1%(v/v), and fermentation degree of 12.66%.

As shown in Table 4, compared with the beer fermented by strain US05, strain B21 produced a 14% increase in isobutyl acetate concentration. The concentrations of isoamyl acetate and ethyl caprate were 786.96 μg/L and 230.02 μg/L, respectively. While the beer fermented with the US05, produced only 178.13 μg/L and 24.56 μg/L of these two substances, indicating the majority of isoamyl acetate and ethyl caprate were produced by strain B21. This strain was also capable of producing phenylethyl acetate (225.06 μg/L) and ethyl hexanoate (26.19 μg/L), which were not detected in LA01 fermented beer. Compared with US05 (12.24 mg/L) and LA01 (10.3 mg/L), B21 produced a total ester level of 22.21 mg/L, an increase of about 50%. B21 has a stronger ability to metabolize esters and could bring rich fruit, rose, butter, berry and other aromas to beer (Roberts et al., 2023). Geraniol in beer is converted into β-citronellol and linalool, and linalool and nerol are isomerized into α-terpinol (Dietz et al., 2021). β-citronellol and nerol were detected in B21 fermented beer, which may have been lost during the fermentation process. The concentration of α-terpinol in beer fermented by B21 was 95.26 μg/L, but it was not detected in beer fermented by US05 and LA01, indicating the increased α-terpinol in beer might all be derived from the transformation of linalool by B21. In addition, B21 can also produce 270.52 ng/L of 3MH. The concentrations of 4MMP and 3MHA were 4.16

ng/L and 9.75 ng/L, respectively, and were not detected in the beer fermented by LA01, indicating the increased amount was entirely produced by strain B21, which has high β-lyase activity. The beer fermented by B21 features a prominent tropical fruit aroma, which was associated with the high-concentration thiols produced in it. In addition, a relatively prominent nutty aroma added to the complexity (Edgar Herkenhoff et al., 2024) of the B21 beers.

In addition, the FAN content of B21 and M3 beers was also measured. Compared with the wort control (FAN 257.6 mg/L), the FAN content of B21 beer was 221.6 mg/L, and the FAN content of M3 beer was 190.0 mg/L. This indicates that yeast consumes free amino nitrogen during the fermentation process, which is used for its own growth, reproduction, and synthesis of metabolic products, including the synthesis of required proteins, nucleic acids, or converted into flavor compounds such as higher alcohols and esters by yeast. The FAN content in M3 beer is lower than that in B21, indicating that M3 has a stronger ability to utilize free amino nitrogen. In addition, the concentration of FAN can affect the release of volatile thiols and esters. Ronald Samia et al. (Samia et al., 2025) found that low FAN promotes thiol release (by a factor of three), while medium to high FAN enhances tropical fruit aroma but may cause off flavors such as diacetyl. This indicates that by selecting yeast strains and determining appropriate wort nitrogen sources, the flavor potential of volatile thiols can be stimulated while avoiding off flavors and improving the quality of beer.

4. Discussion

Recently, non-*Saccharomyces* yeasts were increasingly utilized for their ability to enhance aromatic complexity and flavor richness in beer. Species such as *Torulaspora delbrueckii*, *H. uvarum*, and *Lachancea thermotolerans* contribute to diverse sensory profiles through the production of compounds like esters, higher alcohols, and organic acids (Ha et al., 2025; Pirrone et al., 2025). Relatively high enzyme activities such as β-glucosidase and β-lyase make non-*Saccharomyces* yeast beneficial for

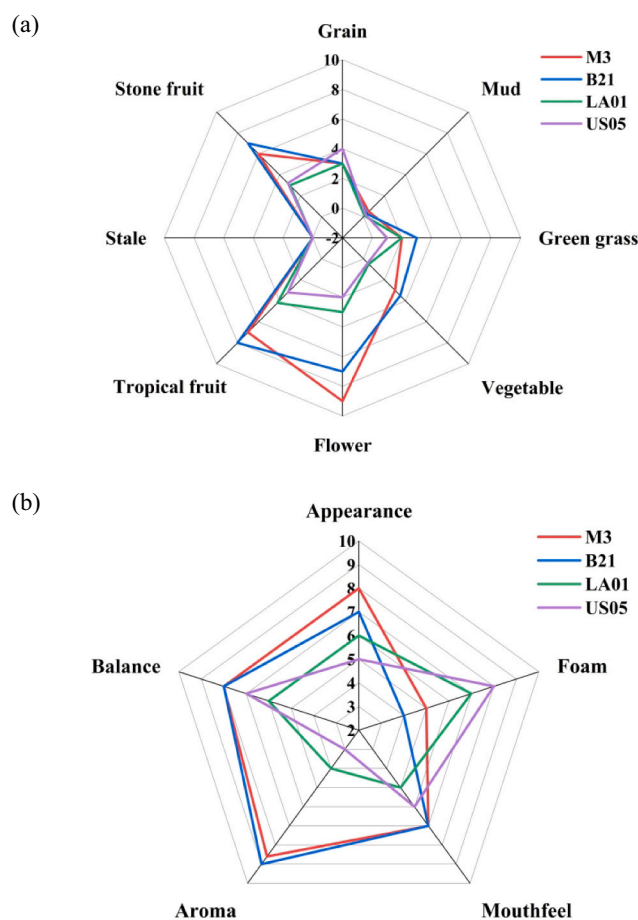


Fig. 6. Sensory evaluation of beers fermented with M3, B21, LA01, and US05: aroma (a), overall (b).

enhancing aroma complexity in fermented beverages (Praia et al., 2022). β -Glucosidase hydrolyzes substrates containing β -D-glucoside bonds, releasing glucose and free glucoside ligands with aromatic properties, promoting the production of aroma compounds. β -Lyase catalyzes the β -elimination of cysteine-S-conjugate compounds, releasing volatile thiols including 4MMP, 3MH and 3MHA, which are known for their tropical aroma in beer (Michel et al., 2019).

Brettanomyces custersii has been reported to hydrolyze glycosides in malt and hops, releasing key aroma compounds such as linalool, methyl salicylate, 1-octen-3-ol, and cis-3-hexen-1-ol in significant quantities during pure or co-fermentation with *Saccharomyces* (Daenen et al., 2008). *Torulaspora delbrueckii* could convert nerol into α -terpineol, while *Kluyveromyces lactis* can transform geraniol into citronellol and linalool, with linalool further converting into α -terpineol (Badura et al., 2025).

Terpenes primarily originate from hops and exist in two forms in hops: free and glycosidically bound. Free terpenes have a low aroma threshold and can impart intense hoppy aromas to beer even at low concentrations, whereas glycosidically bound terpenes are non-volatile and odorless. Three are main pathways for yeast production of terpenoid compounds: de novo synthesis of linalool via the mevalonate pathway, interconversion among linalool, geraniol, and citronellol (Herkenhoff et al., 2024) via linalool isomerase and citronellol dehydrogenase, and hydrolysis of glycosidically-bound aroma compounds by β -glucosidase to release monoterpenes (Holt et al., 2019). M3 increased the contents of linalool, geraniol, β -citronellol, and α -terpineol in the hop matrix by transforming flavor compounds in the hops. It is worth noting that β -citronellol and α -terpineol were not detected in the original hop matrix. Seaton et al. (King & Dickinson, 2000) proposed that

geraniol is converted to β -citronellol by yeast metabolism. But some researchers have suggested the generation of β -citronellol may involve some yet to be described mechanisms. Small amounts of β -pinene, α -humulene, β -dihydroionone β -Caryophyllene, and myrcene were produced in the hop solution after biotransformation. It is suggested the increase in β -citronellol might be partially attributed to the activity of yeast. Compounds like β -pinene, α -humulene, and β -caryophyllene, which are major volatile compounds in hops, are rarely found in beers (Barre et al., 2020). Due to the high hydrophobicity of terpene hydrocarbons, their levels decrease significantly during fermentation through evaporation or absorption into foam (Herkenhoff et al., 2024). Therefore, delaying hop addition time might be effective for retaining terpenes. Strain M3 likely possessed both β -glucosidase and linalool isomerase activities, enabling the transformation of hop compounds during fermentation and resulting in high levels of β -citronellol.

Thiols are divided into free thiols and bound thiols. Free thiols can be directly released into beer through dry feeding to present aroma, while bound thiols require enzyme catalysis to produce free thiols in order to affect flavor and aroma. The bound thiols are mainly precursors of glutathionylation (G-) and cysteinylolation (Cys-), which are formed in grapes and hops through detoxification pathways (Christiaens et al., 2025), and there are also small amounts of precursor substances in malt (Richard et al., 2022). Research has shown yeast biotransformation can increase free thiols by 20%. So far, at least three genes *IRC7*, *STR3*, and *CYS3*, were reported to encode β -lyases in yeast. *IRC7* was key gene in the release of 4MMP from Cys-4MMP and release of 3MH from Cys-3MH (Cordente et al., 2019). However, specific genes regulating 3MH release have not yet been definitively identified, with some researchers suggesting this process might involve the combined effects of multiple genes (Holt et al., 2012). Studies have shown that in most *S. cerevisiae*, the *IRC7* gene contains a 38-bp deletion, resulting in the expression of an inactive *Irc7p* protein lacking β -lyase activity (Cordente et al., 2019; Ruiz et al., 2021; Tofalo et al., 2020). Furthermore, deletions in the *BNA3*, *CYS3*, and *GLO1* genes also led to reduced 4MMP production (Swiegers et al., 2007). Hops contain a large amount of bound thiols, which can be extensively released through conversion by strain B21. After conversion by strain B21, the total thiol concentration increased by 20%, 2.50-fold, and 4.17-fold in the Idaho 7, Mosaic, and Cascade hop matrices, respectively, compared to the original hop matrices. The release of thiols is not only related to the activity of β -lyase, but also to the conditions of the culture medium and temperature. Beer brewing takes place in an acidic and low-temperature environment. Therefore, it is necessary to consider the effects of temperature and pH on enzyme activity to enhance the flavor of beer (Yadav et al., 2022). It is found that the production of 4MMP and 3MH increased when the fermentation temperature raised from 13 °C to 20 °C (Masneuf-Pomarède et al., 2006). The optimal pH value for β -lyase varies depending on the strain. For example, the optimal pH for β -lyase of *K. lactis* was 3.5 and *Schizosaccharomyces pombe* was 4.5–5.5, respectively. The B21 strain exhibited relatively high enzymatic activity within the pH range of 4–6, ensuring its adaptability to the beer fermentation environment and enabling the release of thiols.

The evolution of beer brewing has progressed from spontaneous fermentation to the era of pure-culture fermentation, and now to the modern selective mixed-fermentation brewing stage. The co-fermentation of non-*Saccharomyces* yeasts with brewer's yeasts has contributed to a rich diversity of flavors (Drosou et al., 2025; Herkenhoff et al., 2023). In addition, the ecological interactions between microorganisms in mixed fermentation are currently a research hotspot in aroma driven beer brewing. The two yeast strains isolated from Chinese *Baijiu* fermented grains in this study demonstrate significant potential for application in mixed fermentation, enhancing the floral, fruity, and complexity notes of beer (Edgar Herkenhoff et al., 2024). The idea of designed mixed fermentation using “*Saccharomyces*+ non-*Saccharomyces*/lactic acid bacteria” (Herkenhoff et al., 2023) supports new possibilities for the creation of differentiated, high-quality products.

5. Conclusion

In this study, *W. anomalus* M3 and *P. kluyveri* B21 were screened from fermented grains of *Baijiu*. In a hop matrix, high β -glucosidase strain M3 had advantages in synthesizing linalool, geraniol, β -citronellol, α -terpinol, and ethyl acetate, while high β -lyase strain B21 showed advantages in synthesizing thiols, phenylethyl acetate, and ethyl hexanoate. Applying M3 in beer brewing process significantly improved the esters level of beer imparting flower and fruity flavor to beer. B21 helped improved the thiols compounds in beer, especially 3MH. These non-*Saccharomyces* yeasts made positive contributions to the conversion of esters, terpenes, and thiols in hops and beer fermentation, which could be used as beer aroma enhancing strains. This study offers a technical scheme for the development of new beer products.

CRedit authorship contribution statement

Jinjing Wang: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Xinyue Gu:** Writing – original draft, Methodology, Investigation, Data curation. **Glen Fox:** Writing – review & editing. **Feiyun Zheng:** Visualization, Resources. **Chunfeng Liu:** Visualization. **Chengtuo Niu:** Visualization. **Qi Li:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research was funded by Key Laboratory of Industrial Biotechnology, Ministry of Education, School of Biotechnology, Jiangnan University (KLIB-KF202303) and National Natural Science Foundation of China (No. 31771963).

Data availability

Data will be made available on request.

References

- Allen, T., et al. (2011). Influence of grape-harvesting steps on varietal thiol aromas in sauvignon blanc wines. *Journal of Agricultural and Food Chemistry*.
- Badura, J., et al. (2025). Isolation and characterization of haploid heterothallic beer yeasts. *Applied Microbiology and Biotechnology*, 109(1), 17.
- Barre, B. P., et al. (2020). Intragenic repeat expansion in the cell wall protein gene HPF1 controls yeast chronological aging. *Genome Research*, 30(5), 697–710.
- Basso, R. F., Alcarde, A. R., & Portugal, C. B. (2016). Could non-*Saccharomyces* yeasts contribute on innovative brewing fermentations? *Food Research International*, 86, 112–120.
- Belda, I., et al. (2016). Unraveling the enzymatic basis of wine “Flavorome”: A Phylo-functional study of wine related yeast species. *Frontiers in Microbiology*, 7–2016.
- Christiaens, R., et al. (2025). Evidence in lager yeasts of β -Lyase activity breaking down γ -GluCys-conjugates more efficiently than Cys-conjugates to odorant beer Polyfunctional thiols. *Molecules*, 30(2), 325.
- Cordente, A. G., et al. (2019). Inactivating mutations in *Irc7p* are common in wine yeasts, attenuating carbon-sulfur β -Lyase activity and volatile sulfur compound production. *Applied and Environmental Microbiology*, 85(6), 6.
- Costantini, A., et al. (2025). Exploring the influence of different *Saccharomyces cerevisiae* strains and hop varieties on beer composition and sensory profiles. *Foods*, 14(13).
- Daenen, L., et al. (2008). Screening and evaluation of the glucoside hydrolase activity in *Saccharomyces* and *Brettanomyces* brewing yeasts. *Journal of Applied Microbiology*, 104(2), 478–488.
- Decao, M. A., et al. (2018). Rapid screen of wild yeast with high β -glucosidase activity and evaluation of its glycosidase in winemaking. *Journal of Northwest A & F University(Natural Science Edition)*, 46(1), 129–135.
- Dietz, C., et al. (2021). Exploring the multisensory perception of terpene alcohol and sesquiterpene rich hop extracts in lager style beer. *Food Research International*, 148, Article 110598.
- Drosou, F., et al. (2025). Unlocking the potential of *Zygosaccharomyces bailii*: A novel non-*Saccharomyces* yeast in brewing. *BrewingScience*, 78(5/6), 55–65.
- Edgar Herkenhoff, M., Brödel, O., & Frohme, M. (2024). Aroma component analysis by HS-SPME/GC-MS to characterize lager, ale, and sour beer styles. *Food Research International*, 194, Article 114763.
- Gros, J., Tran, T. T. H., & Collin, S. (2013). Enzymatic release of odourant polyfunctional thiols from cysteine conjugates in hop. *Journal of the Institute of Brewing*, 119(4), 221–227.
- Guan, Y., et al. (2023). *Evaluating the physiology and fermentation performance of the lager yeast during very high gravity brewing with increased temperature* (p. 173). Lwt-Food Science and Technology.
- Ha, S., Qu, T., Zeng, Q. B., Liu, J. J., Zhang, J. Q., Chen, M., & Yang, H. (2025). Research progress on aroma enhancement strategies for fruit wine. *Systems Microbiology and Biomanufacturing*, 5, 1382–1394.
- Han, X., et al. (2023). Application of non-*Saccharomyces* yeasts with high β -glucosidase activity to enhance terpene-related floral flavor in craft beer. *Food Chemistry*, 404, Article 134726.
- Herkenhoff, M. E., Brödel, O., & Frohme, M. (2024). Hops across continents: Exploring how terroir transforms the aromatic profiles of five hop (*Humulus lupulus*) varieties grown in their countries of origin and in Brazil. *Plants*, 13(19), 2675.
- Herkenhoff, M. E., et al. (2023). The combination of omics strategies to evaluate starter and probiotic strains in the Catharina sour Brazilian-style beer. *Food Research International*, 167, Article 112704.
- Holt, S., Cordente, A. G., & Curtin, C. (2012). *Saccharomyces cerevisiae* STR3 and yeast cystathionine β -lyase enzymes: The potential for engineering increased flavor release. *Bioengineered*, 3(3), 180–182.
- Holt, S., et al. (2019). The molecular biology of fruity and floral aromas in beer and other alcoholic beverages. *FEMS Microbiology Reviews*, 43(3), 193–222.
- Karauli, J., et al. (2024). Bioprospecting of *Metschnikowia pulcherrima* strains, isolated from a vineyard ecosystem, as novel starter cultures for craft beer production. *Fermentation*, 10(10), 513.
- King, A., & Dickinson, J. R. (2000). Biotransformation of monoterpene alcohols by *Saccharomyces cerevisiae*, *Torulaspora delbrueckii* and *Kluyveromyces lactis*. *Yeast*, 16(6), 499–506.
- Klimczak, K., et al. (2024). Application of non-*Saccharomyces* yeast for the production of low-alcohol beer. *Foods*, 13(20), 3214.
- Krogerus, K., Rettberg, N., & Gibson, B. (2023). Increased volatile thiol release during beer fermentation using constructed interspecies yeast hybrids. *European Food Research and Technology*, 249(1), 55–69.
- Larrouque, M. N., et al. (2021). Effect of *Saccharomyces* and non-*Saccharomyces* native yeasts on beer aroma compounds. *International Journal of Food Microbiology*, 337, Article 108953.
- Liang, F., Zhong, Z., & Ban, S. (2025). Variation of flavor substance-producing microbiota in four seasons of Chinese liquor fermentation. *Systems Microbiology and Biomanufacturing*, 5, 1503–1512.
- Liu, Y., et al. (2024). Comparison of polyfunctional thiol, element, and total essential oil contents in 32 hop varieties from different countries. *Food Chemistry*, 455(15), Article 139855.
- Masneuf-Pomarede, I., et al. (2006). Influence of fermentation temperature on volatile thiols concentrations in sauvignon blanc wines. *International Journal of Food Microbiology*, 108(3), 385–390.
- Michel, M., et al. (2019). Screening of brewing yeast β -lyase activity and release of hop volatile thiols from precursors during fermentation. *Brewing Science*, 72(11–12), 179–186.
- Peces-Pérez, R., et al. (2022). Biomodulation of physicochemical parameters, aromas, and sensory profile of craft beers by using non. *ACS Omega*, 7(21), 17822–17840.
- Pirrone, A., et al. (2025). Use of non-conventional yeasts for enhancing the sensory quality of craft beer. *Food Research International*, 208, Article 116164.
- Praia, A. B., et al. (2022). Sour beer with *Lactocaseibacillus paracasei* subsp. *paracasei* F19: Feasibility and influence of supplementation with *Spondias mombin* L. Juice and/or By-Product. *Foods*, 11(24), 4068.
- Richard, W. M., et al. (2022). The sensorial and chemical changes in beer brewed with yeast genetically modified to release Polyfunctional thiols from malt and hops. *Fermentation*, 8(8), 370.
- Roberts, R., et al. (2023). Online monitoring of higher alcohols and esters throughout beer fermentation by commercial *Saccharomyces cerevisiae* and *Saccharomyces pastorianus* yeast. *Journal of Mass Spectrometry*, 58(10), 4959.
- Ruiz, J., et al. (2021). Global distribution of *IRC7* alleles in *Saccharomyces cerevisiae* populations: A genomic and phenotypic survey within the wine clade. *Environmental Microbiology*, 23(6), 3182–3195.
- Samia, R., Shayevitz, A., Fischborn, T., & Shellhammer, T. H. (2025). Wort nitrogen and yeast strain drive thiol release, flavor expression, and fermentation performance in beer. *Journal of the American Society of Brewing Chemists*, 1–14.
- Serra Colomer, M., et al. (2020). Biotransformation of hop derived compounds by *Brettanomyces* yeast strains. *Journal of the Institute of Brewing*, 126(3), 280–288.
- Swiegers, J. H., et al. (2007). Engineering volatile thiol release in *Saccharomyces cerevisiae* for improved wine aroma. *Yeast*, 24(7), 561–574.
- Takoi, K. D. M., Shinkaruk, S., Thibon, C., & Tominaga, T. (2009). Specific flavor compounds derived from Nelson Sauvin hop and synergy of these compounds. *BrewingScience*, 62, 108–118.
- Tofalo, R., et al. (2020). Correlation between *IRC7* gene expression and 4-mercapto-4-methylpentan-2-one production in *Saccharomyces cerevisiae* strains. *Yeast*, 37(9), 487–495.
- Wang, X., et al. (2025). Construction of synthetic microbial community for pyrazines metabolism in sauce-flavor *Baijiu*. *Systems Microbiology and Biomanufacturing*, 5(2), 679–691.
- Xiaohong, T., et al. (2020). Research progress on beta-glucosidase-producing yeast in the brewing process of wine. *China Brewing*, 39(4), 7–12.

Yadav, D. K., Chand, K., & Kumari, P. (2022). Effect of fermentation parameters on physicochemical and sensory properties of burans wine. *Systems Microbiology and Biomanufacturing*, 2, 380–392.

Zheng, F., et al. (2012). Isolation and identification of a beta-glucosidase-producing strain and enzymatic characteristics of the beta-glucosidase. *Microbiology*, 39(8), 1059–1068.

Zott, K., et al. (2011). The grape must non-Saccharomyces microbial community: Impact on volatile thiol release. *International Journal of Food Microbiology*, 151(2), 2.